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Physics

Standard level

Paper 1

7 November 2024

Zone A afternoon | **Zone B** afternoon | **Zone C** afternoon

45 minutes

Instructions to candidates

- Do not open this examination paper until instructed to do so.
- Answer all the questions.
- For each question, choose the answer you consider to be the best and indicate your choice on the answer sheet provided.
- A clean copy of the **physics data booklet** is required for this paper.
- The maximum mark for this examination paper is **[30 marks]**.

1. The radius of a sphere is R . The absolute uncertainty in R is ΔR .

What is the fractional uncertainty in the volume of the sphere?

A. $\frac{3\Delta R}{R}$

B. $\left(\frac{\Delta R}{R}\right)^3$

C. $\frac{4\pi\Delta R}{3R}$

D. $4\pi\left(\frac{\Delta R}{R}\right)^3$

2. Which quantity has the fundamental SI units of kg s^{-3} ?

A. Power

B. Intensity

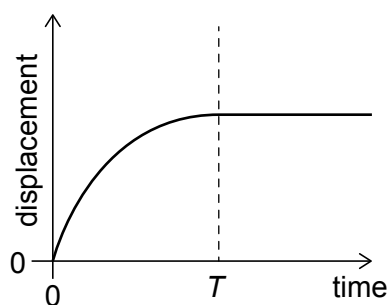
C. Momentum

D. Energy

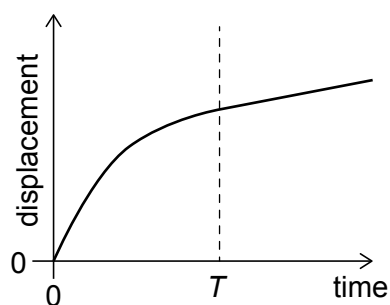
3. A runner accelerates from rest and reaches a constant speed at time T .

What is the variation of displacement with time?

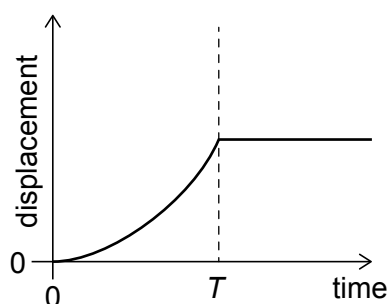
A.



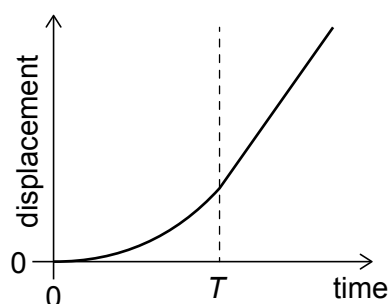
B.



C.



D.

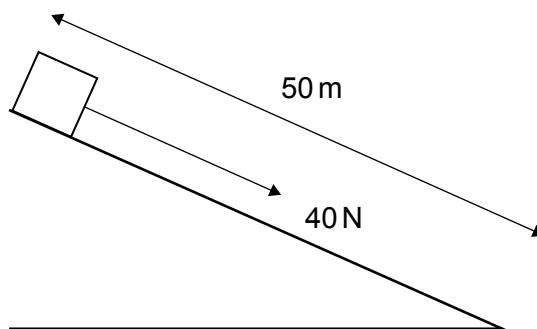


4. A balloon rises at a steady vertical velocity of 10 m s^{-1} . An object is released at rest relative to the balloon when the object is at a height 120 m above the ground. Air resistance is negligible.

What time, to the nearest second, does the object take to hit the ground?

- A. 4 s
- B. 5 s
- C. 6 s
- D. 12 s

5. An object of mass 10 kg is initially at rest at the top of a slope of length 50 m. A constant **net** force of 40 N acts on the object down the slope.



What is the speed of the object at the bottom of the slope?

- A. 20 ms^{-1}
 - B. 40 ms^{-1}
 - C. 200 ms^{-1}
 - D. 400 ms^{-1}
6. When a horizontal spring is extended by a distance x , the force acting on the spring is F . The extension of the spring is directly proportional to the force.

What is the energy stored in the spring?

- A. $\frac{F}{2x}$
- B. Fx
- C. $\frac{Fx}{2}$
- D. $\frac{Fx^2}{2}$

7. An object of mass 6.0 kg is accelerated horizontally from rest in 4.0 s to reach a speed of 10 m s^{-1} .

What is the average power delivered to the object?

- A. 7.5 W
- B. 15 W
- C. 75 W
- D. 150 W

8. An athlete of mass M jumps vertically upward, lifting their centre of mass a distance h after losing contact with the ground.

What is the impulse that acts on the athlete during contact with the ground?

- A. Mgh
- B. $\frac{M}{gh}$
- C. $M\sqrt{2gh}$
- D. $\frac{M}{\sqrt{2}}\sqrt{gh}$

9. A toy car collides with an identical toy car at rest. The cars stick together. The surface is frictionless.

Three statements about this collision are:

- I. The speed of one car decreases and the speed of the other car increases.
- II. Total momentum is conserved.
- III. Kinetic energy is conserved.

Which statements are true?

- A. I and II only
- B. I and III only
- C. II and III only
- D. I, II and III

10. A heater of power P provides energy to a liquid at its boiling point. A mass m of the liquid vaporizes in time t . The specific latent heat of vaporization is L . How much energy is transferred to the surroundings in time t ?

- A. $Pt + mL$
- B. $Pt - mL$
- C. $\frac{mL}{Pt}$
- D. $\frac{Pt}{mL}$

11. The molar mass of helium is 4.0 g mol^{-1} . A sample of helium has mass 0.50 g .

How many molecules of helium are there in the sample?

- A. 3.0×10^{22}
- B. 7.5×10^{22}
- C. 3.0×10^{23}
- D. 7.5×10^{23}

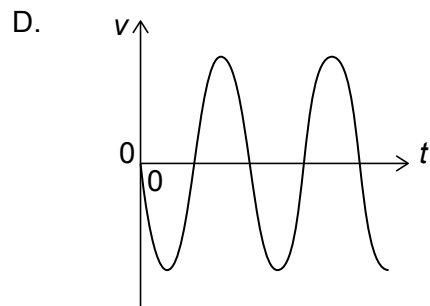
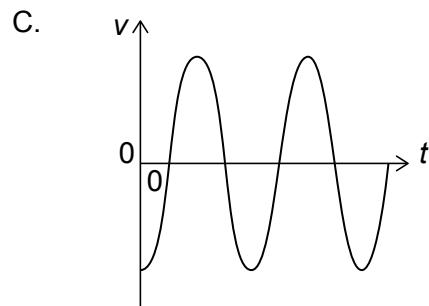
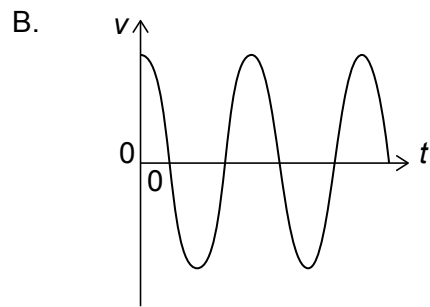
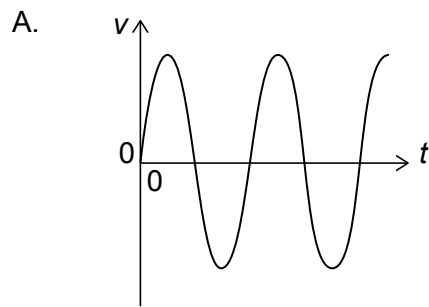
12. An ideal gas is in a container with a movable piston. The piston compresses the gas rapidly.

The temperature of the gas increases because

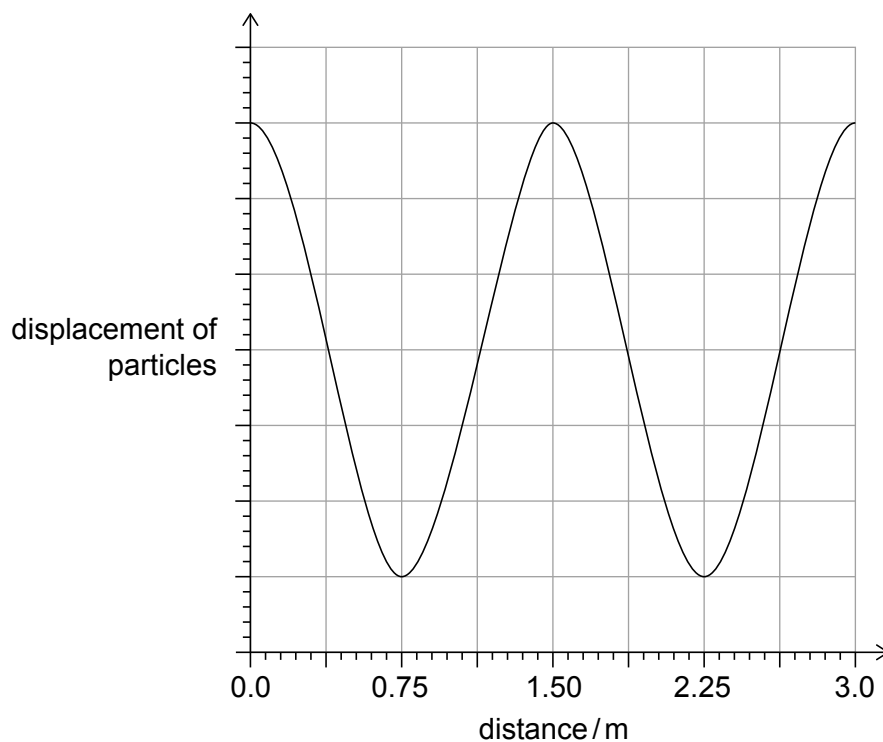
- A. molecules bounce off the piston with increased speed.
- B. molecules collide with the piston more frequently.
- C. molecules exert a greater force on the piston.
- D. the number of molecules per cubic metre has increased.

13. A particle is displaced in the negative direction and released from rest at time $t = 0$. It performs simple harmonic motion (shm).

Which graph shows the variation with time t of the velocity v of the particle?



14. A travelling wave of time period 2.0 s is moving through a medium. The graph shows, for one instant, the variation with distance of the displacement of particles in the medium.

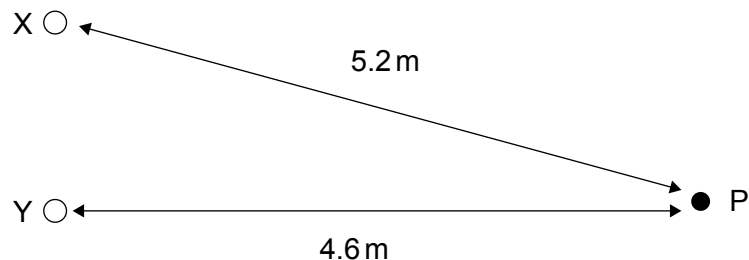


What is the speed of the wave?

- A. 0.75 ms^{-1}
- B. 1.5 ms^{-1}
- C. 3.0 ms^{-1}
- D. 6.0 ms^{-1}

15. Source X and source Y both emit waves of wavelength 0.40 m. There is a constant phase difference of π rad between the sources. Point P is 5.2 m from X and 4.6 m from Y. The amplitude of each wave at P is A .

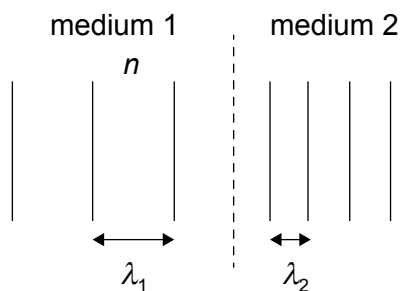
diagram not to scale



What is the amplitude of the resultant wave at P?

- A. 0
- B. A
- C. $\frac{3}{2}A$
- D. $2A$

16. An electromagnetic wave travels from medium 1 to medium 2. The wavelength of the wave in medium 1 is λ_1 . The wavelength of the wave in medium 2 is λ_2 . The refractive index of medium 1 is n .

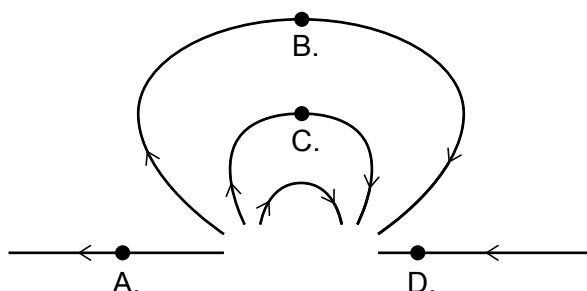


What is the refractive index of medium 2?

- A. $\frac{\lambda_1 n}{\lambda_2}$
- B. $\frac{\lambda_2 n}{\lambda_1}$
- C. $\frac{\lambda_2}{\lambda_1}$
- D. $\frac{\lambda_1}{\lambda_2}$
17. A standing wave is set up in a pipe with one closed end. One harmonic of the standing wave has a frequency 180 Hz. The next harmonic has a frequency of 300 Hz. What is the frequency of the first harmonic of the standing wave?
- A. 30 Hz
- B. 60 Hz
- C. 90 Hz
- D. 120 Hz

18. The diagram shows electrostatic field lines. The source of the field is not shown.

At which position in the field would a negative point charge experience the greatest force to the right?

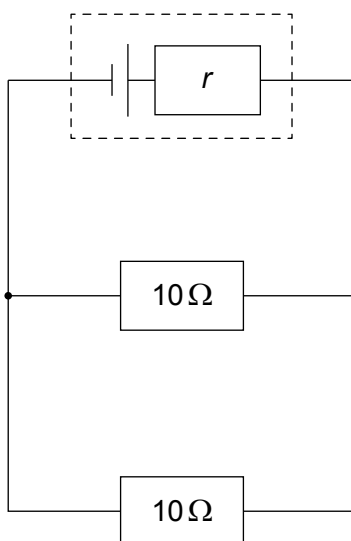


19. A conductor has a resistance of $1.0\text{ k}\Omega$. The length of the conductor is 20 km and the cross-sectional area is 1.0 mm^2 .

What is the resistivity of this conductor?

- A. $2.0 \times 10^{-5}\text{ }\Omega\text{ m}$
- B. $5.0 \times 10^{-5}\text{ }\Omega\text{ m}$
- C. $2.0 \times 10^{-8}\text{ }\Omega\text{ m}$
- D. $5.0 \times 10^{-8}\text{ }\Omega\text{ m}$

20. A cell has an emf of 17.0 V and internal resistance r . It is connected to two $10\ \Omega$ external resistors.

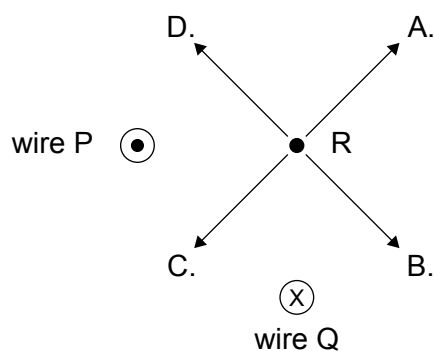


There is a current of 1.5 A in one of the external resistors.

What is the value of r ?

- A. $0.34\ \Omega$
 - B. $0.50\ \Omega$
 - C. $0.67\ \Omega$
 - D. $1.3\ \Omega$
21. P and Q are two parallel wires perpendicular to the page that carry currents of equal magnitude in opposite directions. The current in P is out of the page. R is a fixed point equidistant from P and Q.

What is the direction of the magnetic field produced at R?



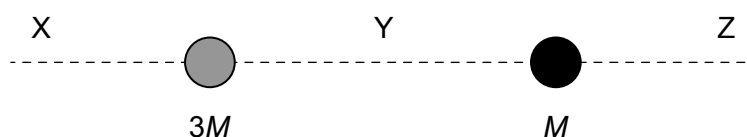
22. X and Y are two points on a disk of radius Z rotating at a constant angular speed.

X is at a distance $\frac{Z}{2}$ from the centre and Y is at a distance Z from the centre.

What is $\frac{\text{linear velocity of X}}{\text{linear velocity of Y}}$ and what is $\frac{\text{acceleration of X}}{\text{acceleration of Y}}$?

	$\frac{\text{linear velocity of X}}{\text{linear velocity of Y}}$	$\frac{\text{acceleration of X}}{\text{acceleration of Y}}$
A.	$\frac{1}{2}$	2
B.	$\frac{1}{2}$	$\frac{1}{2}$
C.	1	2
D.	1	$\frac{1}{2}$

23. The diagram shows two masses that are fixed in position. One has a mass of M and the other has a mass of $3M$. The line through the masses is divided into three segments X, Y and Z.



The net gravitational field can be directed to the left in

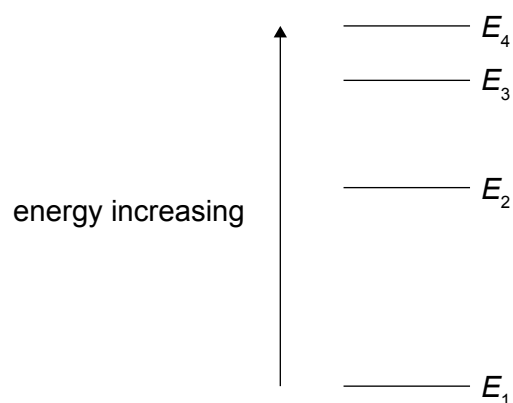
- A. only segment Z.
- B. only segment X.
- C. segments Z and Y.
- D. segments X and Y.

24. An unstable nuclide has too many protons.

What is the most likely decay of the nuclide?

- A. Alpha
- B. Beta minus
- C. Beta plus
- D. Gamma

25. The four energy states for an atom are shown.



What is the lowest frequency of radiation that can be absorbed by the atom?

- A. $\frac{E_1 - E_4}{h}$
- B. $\frac{E_4 - E_1}{h}$
- C. $\frac{E_3 - E_4}{h}$
- D. $\frac{E_4 - E_3}{h}$

26. Which list correctly shows three fundamental forces in increasing order of range?

- A. Weak nuclear, strong nuclear, electromagnetic
- B. Strong nuclear, electromagnetic, weak nuclear
- C. Strong nuclear, weak nuclear, electromagnetic
- D. Electromagnetic, weak nuclear, strong nuclear

27. The reaction $\pi^0 + p \rightarrow p + e^- + \nu_e$ is proposed.

Three conservation laws in particle reactions are:

- I. conservation of charge.
- II. conservation of baryon number.
- III. conservation of lepton number.

Which conservation laws are violated in the proposed reaction?

- A. I and II only
- B. I and III only
- C. II and III only
- D. I, II and III

28. Once research is completed, scientists gather support for the validity of their work by

- A. using common terminology.
- B. collaborating with others.
- C. improving their instrumentation.
- D. obtaining a review by independent scientists.

29. Three energy sources for power stations are:

- I. nuclear fuel.
- II. sunlight.
- III. fossil fuel.

Which energy sources are primary sources?

- A. I and II only
- B. I and III only
- C. II and III only
- D. I, II and III

30. The surface of a planet absorbs an intensity of 400 W m^{-2} and reflects 100 W m^{-2} . The surface of the planet is at equilibrium at a constant temperature.

What is the albedo of the planet surface and the radiated intensity?

	Albedo of the planet surface	Radiated intensity / W m^{-2}
A.	0.20	300
B.	0.20	400
C.	0.25	300
D.	0.25	400
